# Ibrahim Hassan

Fullstack Developer (.NET, Angular)

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## Professional Summary

Results-driven Software Engineer and **ECPC Finalist** experienced in building real-time platforms, hybrid CQRS architectures, and multi-service systems using **ASP.NET Core**, **Clean Architecture**, and **Angular**, backed by automated testing and **CI/CD pipelines**.

## Professional Experience

**Full Stack .NET Developer** Oct 2025 – Nov 2025
*Laundry Management System* – Freelance *Cairo, Egypt*

- Architected and delivered a 3-layer **ASP.NET Core MVC** application (`Web`, `Core`, `Infrastructure`) using **Entity Framework Core**, **SQL Server**, and **AutoMapper** to manage laundry operations across hotels, apartments, and regions.
- Implemented **role-based access control (RBAC)** using **ASP.NET Identity** across dedicated MVC Areas for **3 roles** (Admin, Owner, Employee), securing authentication, authorization, and custom **Hangfire** dashboard filters.
- Engineered **10+ business services** covering dynamic hotel/apartment pricing, invoice generation and repricing, debtor tracking, and categorized expense analytics.
- Automated daily, monthly, and custom-range invoice workflows using **4 Hangfire background jobs** (`Hangfire.Console`), generating **Rotativa PDF** invoices and **ClosedXML Excel** financial reports.
- Built responsive bilingual (**Arabic RTL / English**) Razor views styled with **TailwindCSS** and `.resx` localization, integrating **WhatsApp API** and **MailKit** SMTP for automated customer notifications.

## Technical Projects

**TicketFlow – Enterprise Support Ticket Platform** (Backend | Frontend | Swagger | Live App) 2026

- Architected a 4-layer **Clean Architecture + CQRS** platform in **ASP.NET Core** and **PostgreSQL**, separating write commands (**EF Core**) from read queries (**Dapper** + 11 PL/pgSQL stored functions) to reduce query latency by **60%**.
- Engineered **optimistic concurrency control** via PostgreSQL row-version triggers mapped to **HTTP 409 Conflict** (RFC 7807), alongside MediatR validation and logging pipelines (**FluentValidation**, **Serilog**).
- Secured 20+ RESTful endpoints using **JWT with Refresh Token rotation**, 3-tier RBAC, horizontal customer IDOR data isolation, and fixed-window IP rate limiting (15 requests/sec).
- Achieved a **100%** pass rate across **328 automated tests** (272 unit, 43 API, 13 PostgreSQL integration tests) and automated build, test, and **Docker Compose** workflows via **GitHub Actions CI/CD**.

**Wasla – Smart Microbus Platform** (GitHub | Live Demo) Jan 2026 – Jun 2026

- Developed a localized (**Arabic/English**) real-time transportation platform across 3 Clean Architecture layers, delivering 11 controllers, 16 domain entities, and 86 DTOs in **ASP.NET Core**.
- Engineered **4 concurrent SignalR hubs** pushing live queue positions, driver statuses, GPS coordinates, and route ETAs to 5 user roles across web and mobile clients with sub-second latency.
- Automated atomic trip creation via **AES-GCM encrypted QR** authentication, where a single scan validates tokens, mutates queues, and triggers 4 real-time broadcasts, reducing manual dispatch time by **70%**.
- Optimized GPS tracking using **OSRM routing**, **Haversine filtering**, and **Redis caching** (4-hour TTL), cutting external routing API calls by **65%**; containerized with **Docker Compose**.

**WhatsApp Messaging Infrastructure** (Gateway | Engine | Dashboard) 2026

- Built a 3-service distributed messaging system featuring an **ASP.NET Core orchestration gateway** with SQL Server delivery logging and custom server-to-server authentication filters.
- Engineered a self-healing **headless WhatsApp engine** with a 6-state machine, 45-second heartbeat zombie detection, and exponential backoff recovery, achieving **99%+** session uptime.
- Created a real-time **Angular monitoring dashboard** using Standalone Components and Signals, integrating **SignalR** for live QR streaming, JWT route guards, and auto-reconnect logic.

**E-StoreX – Enterprise E-Commerce Platform** (Backend | Frontend | Swagger | Live Demo) 2025 – Feb 2026

- Built **140+ RESTful endpoints** in a bilingual (**Arabic/English**) Clean Architecture **ASP.NET Core Web API**, orchestrating **Stripe Webhooks**, **Redis caching**, **Hangfire** jobs, and 3 auth providers (JWT, Google & GitHub OAuth).
- Delivered **250+ automated xUnit test cases** using Moq and FluentAssertions, achieving **90%+** code coverage across unit, service, and integration layers.
- Architected 20+ modular **Angular** features using Standalone Components and **Angular Signals**, reducing change detection cycles by **40%** and improving First Contentful Paint (FCP) by **50%** via **SSR** and Lazy Loading.
- Implemented a 4-step **Stripe checkout** flow with full Arabic RTL / English LTR localization, dark/light themes, and persistent cart synchronization.

## Technical Skills

**Languages:** C#, TypeScript, JavaScript, SQL, C++, Java

**Backend (.NET):** ASP.NET Core, Web API, Minimal APIs, MVC, Entity Framework Core, Dapper, MediatR (CQRS), SignalR, Hangfire, LINQ, FluentValidation, AutoMapper, Node.js

**Frontend:** Angular, Angular Signals, RxJS, TypeScript, SSR, TailwindCSS, HTML5, CSS3, SCSS, Localization (i18n / RTL)

**Databases:** PostgreSQL (Stored Functions, Triggers), SQL Server, Redis

**DevOps & CI/CD:** Git, GitHub, GitHub Actions (CI/CD Pipelines), Docker, Docker Compose

**Auth & Security:** JWT, Refresh Token Rotation, OAuth 2.0, ASP.NET Identity, RBAC, AES-GCM Encryption, Rate Limiting

**Testing & Tools:** xUnit, Moq, FluentAssertions, Integration Testing, Serilog, Swagger/OpenAPI, Clean Architecture, REST APIs

**AI Tooling:** Google Antigravity, Claude Code, GitHub Copilot, ChatGPT

**Soft Skills:** Problem Solving, Cross-Functional Team Collaboration, Technical Communication, Time Management, Adaptability, Ownership

## Education

**Bachelor of Information Technology** Sep 2022 – Jun 2026

*Faculty of Science, Minia University* – GPA: 3.4 / 4.0 *Minia, Egypt*

## Achievements & Certifications

- **ECPC Finalist** – Egyptian Collegiate Programming Contest (2024)
- **AAIB Banking Internship** – Arab African International Bank (2025)
- **Professional Certificates (Udemy, 2025):** C# Development · ASP.NET Core Web API · Angular Development

## Languages

Arabic *(Native)* | English